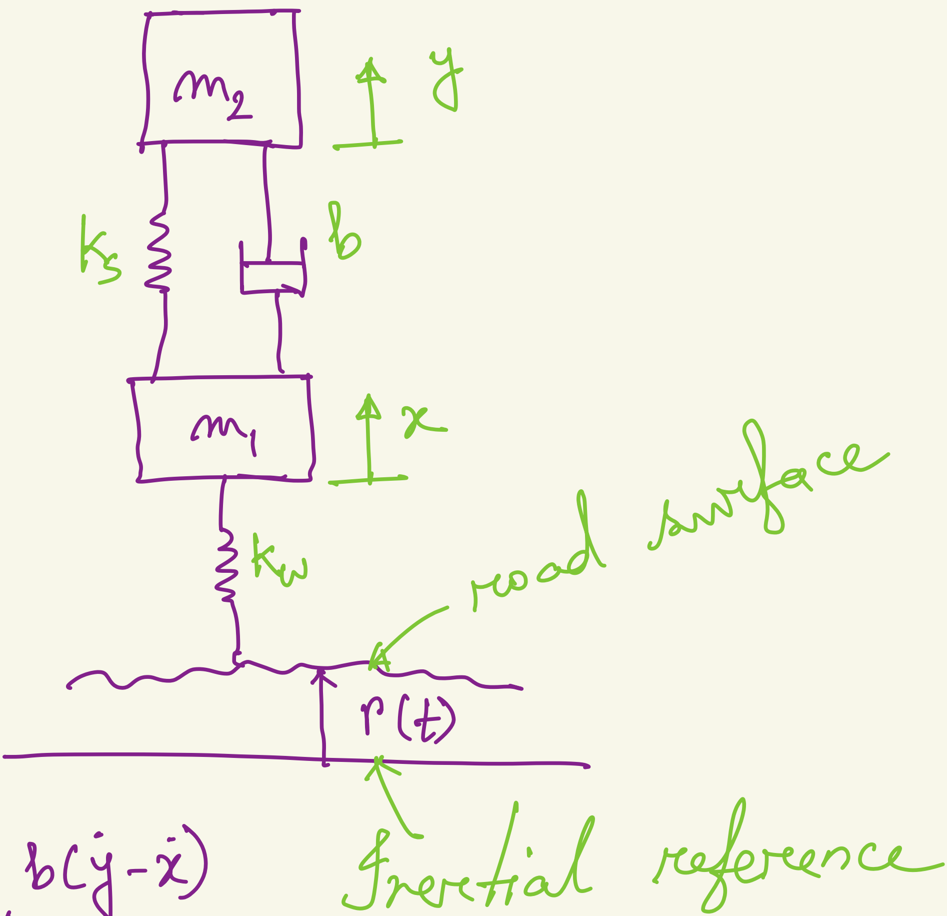
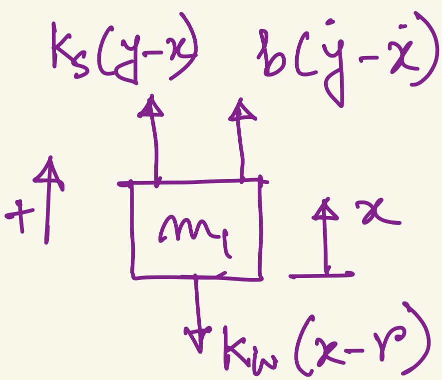
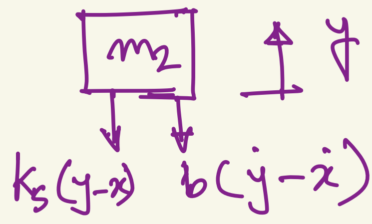
Lecture 9



m_2
y
k_s
b
m_1
x
k_w
road surface
$r(t)$
Inertial reference



$k_s(y-x)$
$b(\dot{y}-\dot{x})$
$+$
m_1
x
$k_w(x-r)$



m_2
y
$k_s(y-x)$
$b(\dot{y}-\dot{x})$

For mass 1:

$$\Sigma F = m_1 \ddot{x}$$

$$\Rightarrow k_s(y-x) + b(\dot{y}-\dot{x}) - k_w(x-r) = m_1 \ddot{x}$$

For mass 2:

$$\Sigma F = m_2 \ddot{y}$$

$$\Rightarrow -k_s(y-x) - b(\dot{y}-\dot{x}) = m_2 \ddot{y}$$

Let us Rearrange and write in the following way after dividing ~~su~~ by individual masses.

$$\boxed{\ddot{x} + \frac{b}{m_1}(\dot{x}-\dot{y}) + \frac{k_s}{m_1}(x-y) + \frac{k_w}{m_1}x} = \frac{k_w}{m_1} r(t)$$

$$\ddot{y} + \frac{b}{m_2}(\dot{y}-\dot{x}) + \frac{k_s}{m_2}(y-x) = 0$$

$$s^2 X(s) + \frac{b}{m_1}\left(s X(s) - s Y(s)\right) + \frac{k_s}{m_1}\left\{X(s) - Y(s)\right\} + \frac{k_w}{m_1} X(s) = \frac{k_w}{m_1} R(s)$$

$$\Big(\overbrace{s^2 + s\frac{b}{m_1} + \frac{k_s}{m_1} + \frac{k_w}{m_1}}^{a_1}\Big) X(s) + \Big(\underbrace{-s\frac{b}{m_1} - \frac{k_s}{m_1}}_{b_1}\Big) Y(s) = \overbrace{\frac{k_w}{m_1}}^{c_1} R(s)$$

$$\underbrace{\left(-s\frac{b}{m_2}-\frac{k_s}{m_2}\right)}_{a_2}X(s)+\overbrace{\left(s^2+s\frac{b}{m_2}+\frac{k_s}{m_2}\right)}^{b_2}Y(s)=0$$

$$a_1X(s)+b_1Y(s)=c_1R(s)$$

$$a_2X(s)+b_2Y(s)=0$$

$$\frac{X(s)}{R(s)} = \frac{\det\begin{bmatrix} c_1 & b_1 \\ 0 & b_2 \end{bmatrix}}{\det\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}}$$

$$\frac{Y(s)}{R(s)} = \frac{\det\begin{bmatrix} a_1 & c_1 \\ a_2 & 0 \end{bmatrix}}{\det\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}}$$

$m_1 = 20$ kg

$m_2 = 375$ kg

$b = 9800$ N.s./m

$$\frac{Y(s)}{R(s)} = \frac{\dfrac{k_w b}{m_1 m_2}\left(s + \dfrac{k_s}{b}\right)}{\left\{ s^4 + \left(\dfrac{b}{m_1} + \dfrac{b}{m_2}\right)s^3 + \left(\dfrac{k_s}{m_1} + \dfrac{k_s}{m_2} + \dfrac{k_w}{m_1}\right)s^2 + \dfrac{k_w b}{m_1 m_2} s + \right.}$$

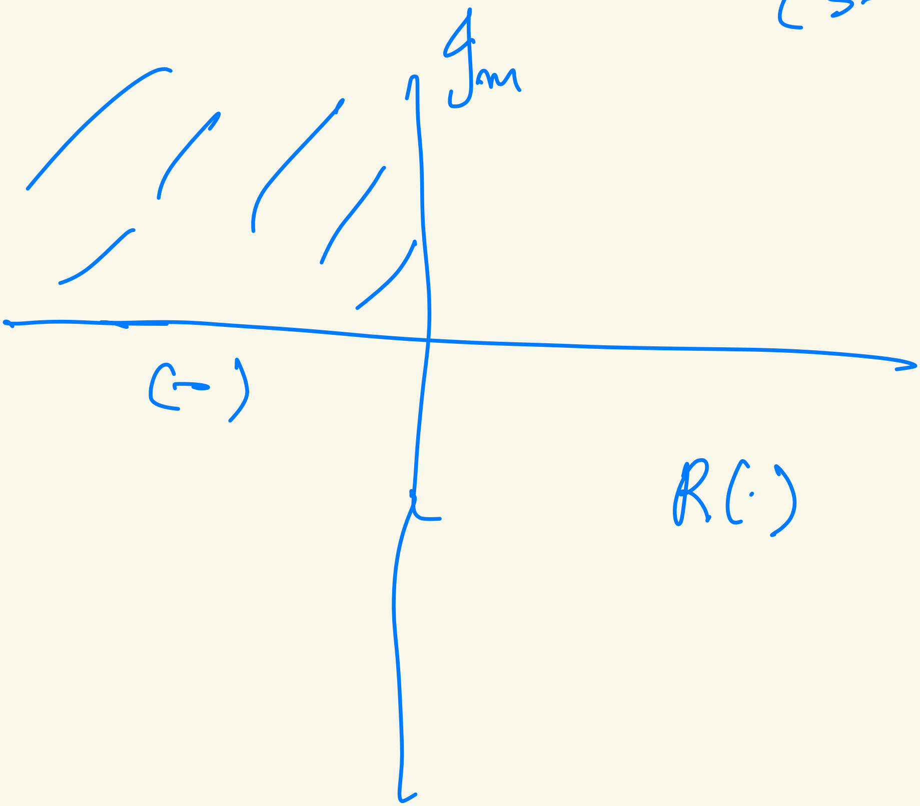
$$\left. \frac{k_w k_s}{m_1 m_2} \right\}$$

$$G(s) = T(s) = \frac{1.3 \times 10^6 (s + 13.3)}{s^4 + (56.1)s^3 + (5.68 \times 10^4)s^2 + 1.3 \times 10^6 s + 1.73 \times 10^6}$$

$$G(s) = \frac{1.3 \times 10^6 (s + 13.3)}{(s - p_1)(s - p_2)(s - p_3)(s - p_4)}$$

poles of the system

$$G(s) = k \cdot \frac{(s - z_1)}{(s-p_1)(s-p_2)(s-p_3)(s-p_4)}$$



$$s = \sigma + j\omega$$

$$G(s) = K \cdot \left\{ \frac{c_1}{s-P_1} + \frac{c_2}{s-P_2} + \frac{c_3}{s-P_3} + \frac{c_4}{s-P_4} \right\}$$

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$$

$$= \frac{1}{s-(-3)}$$

If we take inverse Laplace

$$\mathcal{L}^{-1}\left\{\frac{c_1}{s-P_1}\right\} = c_1 e^{P_1 t}$$

In this way the time function $g(t)$ can be found.

$$g(t) = \sum_{i=1}^{n} c_i e^{p_i t}$$